

ANANT RAI

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Nashik, Maharashtra, India

AI/ML Engineer with experience in computer vision, LLMs, and RAG. I built the MPVE framework (+23 pp OCR accuracy) and trained a YOLOv7 model on 20,327 paediatric X-rays.

Technical Skills

ML / DL Python, NumPy, Pandas, Scikit-learn, YOLOv7/v8, ONNX Runtime, OpenCV, Tesseract OCR, Gradio
LLM / GenAI Ollama (Llama 3.2), Gemini API (1.5 Flash), OpenAI API, RAG pipelines, Prompt Engineering, LangChain basics
CV Research MPVE framework (FVR, GSVE, TCVL), Object Detection, Bounding Box Regression, Attention Heatmaps, GNN
MLOps / Deploy FastAPI, Docker, Vercel, Hugging Face Spaces, Git/GitHub, CI/CD Pipelines, Postman
Data SQLAlchemy, SQLite, MongoDB, REST APIs, Pydantic, JWT, WebSockets
Frontend React.js, Tailwind CSS, JavaScript (ES6+), Chart.js, Jinja2

Experience

Full-Stack Developer Intern (AI Integration Focus)

May 2025 – Aug 2025

Unified Mentor Pvt. Ltd.

Remote

- Integrated Gemini and OpenAI APIs into 2 product features (AI content generation, smart replies); average session length up **40%**.
- Deployed 3 MERN stack apps to Render/Netlify; CI/CD pipelines reduced release time from days to **under 1 hour**.
- Built FastAPI inference endpoints for AI features; page load time down **20%** via code splitting and lazy loading.

Projects

YOLOv7 Paediatric Fracture Detector [GitHub] [HF Demo]

Stack: YOLOv7-p6, ONNX Runtime, OpenCV, Gradio, Python, NumPy, Hugging Face Spaces

Trained YOLOv7-p6 on **20,327 X-rays** (6,091 paediatric cases) to detect 9 fracture/anomaly classes. ONNX export: **2–3x CPU inference speedup**. Confidence-weighted attention heatmaps for radiologist-readable explainability. Live on HF Spaces.

Sanjivani – AI Healthcare Platform [GitHub]

Stack: FastAPI, Python, ONNX YOLOv7, Tesseract, MPVE, Ollama/Llama 3.2:3b, OpenCV, SQLAlchemy, React, Tailwind CSS

End-to-end medical AI system. MPVE-enhanced OCR: **68% to 91%** on 100 samples; **12 Indian languages** via NVIDIA Magpie TTS. Local LLM (Llama 3.2:3b) extracts medicine/dosage offline. Doctor dashboard with YOLOv7 X-ray fracture detection.

ISRO MOSDAC Knowledge Graph AI Bot [Bharatiya Antariksh Hackathon – ISRO 2025]

Stack: Python, LLM, Graph Neural Networks, Hierarchical Multi-Agent RAG, Knowledge Graph of Thoughts (KG-oT)

LLM assistant for ISRO's MOSDAC satellite portal. Spatio-temporal knowledge graph with GNNs for multi-hop reasoning; persistent context via Knowledge Graph of Thoughts (KG-oT).

SkillGenie – AI Career Advisor [GitHub] [Live Demo]

Stack: React, Node.js, Gemini AI 1.5 Flash, YouTube API, Google Search API, JWT, Vercel

Gemini 1.5 Flash generates personalised learning roadmaps, skill-gap analysis, and 5–8 year market projections. Curated resources via YouTube/Google APIs; 24/7 AI mentor chat with context-aware responses.

Research

[R1] MPVE: Multi-Perspective Vision Enhancement for Computer Vision and Medical Intelligence

- Developed a 3-module CV framework: FVR, GSVE, and TCVL.
- Improved Tesseract OCR accuracy on handwritten prescriptions from **68% to 91% (+23 pp)**.
- Awarded **1st Prize at SRPS 2025** and presented at conference level.

[R2] Integrating OR Methodologies: Enhancing Efficiency and Risk Mitigation Across Healthcare [View Paper]

- Published in **IEJSE Vol. 8, No. 9 (2025)** and secured **1st Position** in conference presentation.
- Designed a hybrid OR-ERP model using LP, MIP, DES, Monte Carlo, and DEA techniques.
- Optimized ICU bed allocation, patient flow, and ML-based healthcare forecasting.

Education

Bachelor of Science – Computer Science

Jun 2024 – May 2028

Ashoka Center for Business and Computer Studies, Savitribai Phule Pune University (SPPU), Nashik

SGPA: **9.73** (Sem 1) — **9.64** (Sem 2) — 44 of 44 credits completed

Higher Secondary Certificate (HSC) – Science Stream

Jun 2022 – Mar 2024

HPT Arts and RYK Science College, Nashik — Maharashtra State Board

Total: 356/600 — Grade A

Achievements

- International Publication:** IEJSE Vol. 8, No. 9, Sep 2025 – OR+ML hospital efficiency model [View].
- 5 Hackathons:** ISRO Bharatiya Antariksh (LLM + knowledge graph); GHRhack 2.0 (prescription AI, 30 hrs); SunHacks (StudyGenie, 36 hrs, UX award); Gen AI Exchange / Google Cloud; Avishkar 2025 SPPU Zonal.
- Research Cell Appreciation – ACBCS (Mar 2026):** Coordinated Avishkar, SRPS, and IPR workshops.

Certifications

NumPy Mastery: Numerical Computing – Udemy 2024 **Data Science Essentials** – Udemy (UC-758c55c6) Mar 2025 **Google Cloud AI Lab** – GDG / HackwithIndia 2025